

Gesture-Controlled AR Robotic Puzzle for Computational Thinking

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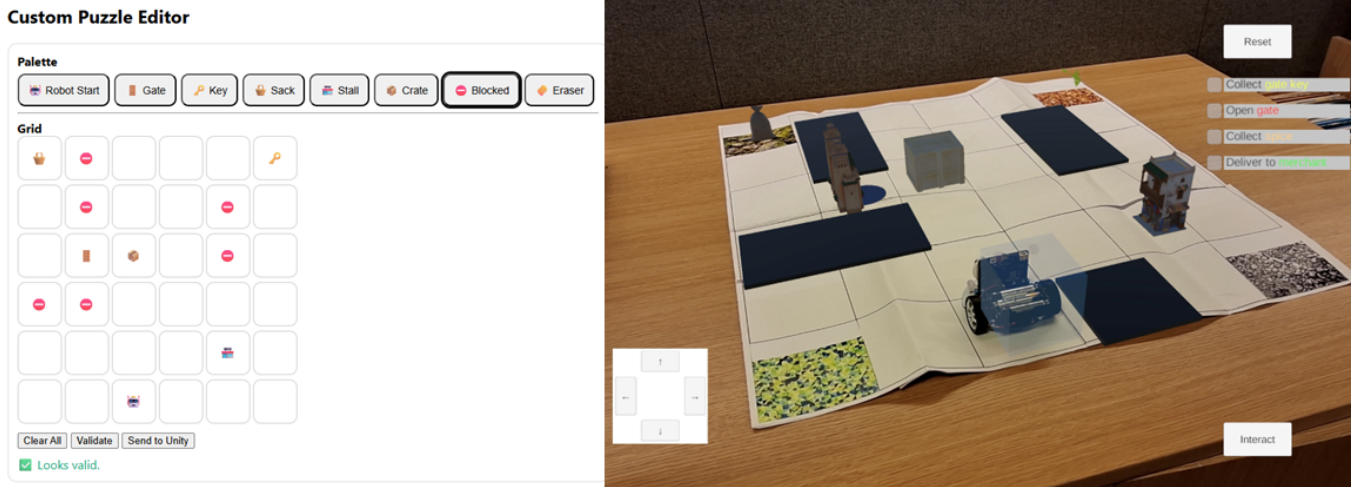


Figure 1: The user scenario of our design: A learner creates a custom puzzle using a web-based editor (left), which is then loaded and visualized in an augmented reality puzzle environment synchronized with a physical robot navigating a grid-based board (right). This mixed-reality setup tightly couples puzzle authoring with physical execution, enabling learners to engage with computational thinking concepts through both design and embodied interaction.

Abstract

Computational thinking (CT) is a central goal in K–12 education, yet learners often struggle to connect abstract programs to physical consequences. We present a gesture-controlled augmented reality (AR) robotic puzzle environment that supports multi-modal programming over a shared computational state. A mobile robot navigates a physical board spatially registered to an AR puzzle world on a tablet. We conducted a formative study with two participants (an HCI professor and a computer science student) to examine usability and interaction design considerations. Results indicate that the sequencer supported efficient planning and reflection, whereas gesture-based control introduced reliability and feedback challenges, including recognition uncertainty and unclear state-to-action confirmation. Participants found puzzle authoring engaging but reported difficulty constructing meaningful challenges, suggesting a need for scaffolding. This work contributes a mixed-reality robotic puzzle platform that enables transitions between symbolic

sequencing, embodied control, and learner-authored environment design, highlighting key requirements for supporting CT learning.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Mixed / augmented reality**; **Gestural input**; *Empirical studies in HCI*; *User interface design*; • **Applied computing** → **Education**; **Interactive learning environments**.

Keywords

computational thinking, educational robotics, puzzle authoring, embodied interaction, multi-modal programming, constructionist learning, programming education

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1 Introduction

Computational thinking (CT) has become a core learning objective in K–12 education, encompassing skills such as sequencing, abstraction, decomposition, conditionals, and debugging. [3, 11]



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Many CT learning tools introduce these concepts through screen-based programming environments or educational robotics kits, yet learners often struggle to connect abstract instructions with their physical consequences [11]. Prior work in tangible, embodied, and mixed-reality learning suggests that grounding computation in physical action can improve engagement and understanding [4]; however, most existing systems either restrict learners to a single programming modality or cast them primarily as solvers of predefined problems, limiting opportunities to **experience** execution as an embodied, inspectable process [1, 4].

In this paper, we present an augmented reality (AR) robotic learning environment that supports multi-modal programming interaction over a shared computational state, explicitly foregrounding **hand-gesture control** as a bridge between symbolic plans and situated action. A physical robot moves across a 6×6 board that is spatially aligned with an AR puzzle environment displayed on a tablet. Learners interact through a web-based interface that integrates: 1) a command sequencer for planned, stepwise programs, 2) hand-gesture control for immediate, embodied movement and step enactment, and 3) a puzzle editor for authoring tasks. This integration enables learners to fluidly shift between composing a program, gesturally executing and inspecting robot behavior in context, and revising either the program or the puzzle without switching systems. Beyond solving predefined puzzles, learners can author their own computational environments by designing layouts, defining goals, and introducing constraints—engaging CT through abstraction and decomposition at the level of problem construction rather than solely at program execution.

We conducted a formative user study with two participants, an HCI professor and a computer science student, to examine usability and interaction design considerations for this multi-modal workflow. The study indicated that the sequencer and puzzle-authoring features supported efficient planning and reflection, while gesture-based control introduced distinct usability and reliability challenges—especially around orientation, discoverability, and the need for clear, immediate feedback linking a gesture to robot intent and system state. These observations underscore that gesture input can meaningfully support embodied exploration and debugging, but only when paired with robust recognition, explicit state cues, and interaction techniques that reduce ambiguity.

This work contributes a mixed-reality robotic learning system that integrates multiple programming modalities—most notably hand-gesture control and symbolic sequencing—within a shared computational framework, and positions puzzle design as a CT activity. Collectively, these contributions illustrate how gesture-enabled embodied execution, combined with learner-authored problem spaces, can support more holistic approaches to CT education.

2 Related Work

Our work is informed by prior research in educational robotics and CT, tangible and embodied interaction, hand-gesture interaction for robot control, and AR learning environments. We review these areas to position our system and identify gaps in how existing approaches support programming modalities, embodied control, and learner agency.

Educational Robotics and Computational Thinking: Educational robotics systems are widely used to introduce CT concepts such as sequencing, debugging, and algorithmic reasoning [1]. Many platforms rely on block-based or text-based programming interfaces that allow learners to construct programs and observe robot behavior. These systems have been shown to support engagement and learning, particularly when tasks are structured and goal-oriented [1]. However, such approaches often emphasize symbolic program construction and may obscure the relationship between abstract instructions and physical execution, especially for novice learners [4]. Additionally, most robotics-based CT tools position learners primarily as problem solvers, working within predefined environments rather than shaping the computational problems themselves [1, 12].

Augmented Reality in Educational and Robotic Systems: AR has been explored as a means to visualize abstract processes, provide contextual feedback, and support situated learning. In educational robotics, AR is often used to overlay robot state, paths, trajectories, or instructional cues to support real-time control and understanding of robot behavior [8]. Recent systems have also explored AR-based authoring interfaces for physical robotic artifacts, allowing users to design interactive games or experiences that are executed by a physical robot, such as AR car game authoring environments built on Arduino platforms [2]. However, these systems typically focus on single interaction modalities and do not integrate multiple programming approaches within a shared computational framework. Furthermore, they typically focus either on visualization or content authoring, rather than supporting integrated programming workflows [5]. Moreover, AR-based learning systems often treat learners as consumers of predefined scenarios, limiting their ability to design or modify the computational environment itself [8].

Hand-Gesture Interaction for Robot Control and AR: Hand-gesture input has a long history in human-robot interaction as a means of issuing commands through familiar bodily actions, especially for non-expert users. Gesture-control research shows strong potential for intuitive robot command input [6, 7, 9]. Although constructionist approaches emphasize learning through making and learner-authored artifacts [10, 12], most educational computing systems focus on constructing programs rather than enabling learners to construct computational environments [1, 4, 12]. Addressing these gaps requires systems that integrate multiple programming modalities (including gesture control) while supporting learner-authored environments within a single coherent experience. To address these gaps, we explore the following research questions:

- **RQ1:** How do learners adopt and transition between gesture-based robot control and symbolic sequencing when interacting with a gesture-controlled AR robotic puzzle? This question examines how each modality supports planning, execution, and debugging, and what interaction breakdowns or handoffs occur between direct control and programmed control.
- **RQ2:** How does learner-authored puzzle design in an AR robotic environment shape CT practices of abstraction and decomposition? This question investigates how designing puzzle layouts, goals, and constraints influences learners'

reasoning about problem structure, strategy formation, and iterative refinement.

3 Concept Idea

Design Rationale: The system’s design is guided by two core principles: 1) CT involves multiple forms of reasoning, including planning, execution, and reflection, which are often separated across different tools or interaction styles. We aim to unify these processes by supporting multiple programming modalities, sequencing, and embodied control within a single, coherent environment that operates over a shared computational state. 2) Prior research suggests that learners benefit not only from solving problems but also from designing the problems themselves, allowing them to externalize computational logic through the structure of the environment.

Iterative Design Process: The concept emerged through an iterative design process informed by related work, internal prototyping, and design critique. As seen in Table 1, each iteration addressed limitations identified in earlier prototypes, ultimately leading to a unified system that integrates physical interaction, AR visualization, multi-modal programming, and puzzle authoring.

Prototype Overview: It has three integrated components: a physical robot and board, an AR puzzle environment, and a web-based interaction interface. As seen in Figure 1, a small mobile robot moves across a 6×6 physical board, while a tablet displays an AR puzzle world that is spatially registered to the board. Learners interact with the robot through a web-based interface running on a laptop. This interface supports two complementary interaction modes: 1) a command sequencer that allows learners to compose and execute step-by-step instructions, and 2) a hand gesture control mode that enables direct, embodied control of the robot’s movement and actions. Both modalities operate on the same underlying puzzle and robot state, allowing learners to switch fluidly between planning and execution.

Learner-Authored Puzzle Design: Learners can create their own puzzles using a web-based editor. The editor allows users to design custom layouts by placing obstacles, goals, and interactive elements on the grid. By authoring puzzles, learners engage in CT through abstraction and decomposition, as they must consider how environmental constraints and goals influence the sequence of actions required to complete the task.

Learning Context: The system is intended for K–12 learners and supports a range of CT concepts, including sequencing, abstraction, decomposition, conditionals, and debugging. Teachers may select predefined puzzles to introduce concepts or allow learners to design their own challenges, enabling flexible classroom integration. By combining embodied interaction, symbolic programming, and learner-authored environments, the concept aims to support a more holistic and engaging approach to computational thinking education.

4 User Study

We conducted a formative user study to explore how users interact with the system’s multi-modal programming interfaces and learner-authored puzzle design workflow. The goal of the study was to identify usability issues, interaction breakdowns, and early design insights rather than to measure learning outcomes.

Participants: Two participants took part in the study, selected to provide contrasting perspectives: 1) P1 was a university professor with extensive experience in educational technology and prior exposure to robotics, AR, and similar interactive systems. 2) P2 was a computer science undergraduate student familiar with programming concepts but with limited experience using physical or AR robotic systems. This combination enabled us to capture feedback from both an expert evaluator and an advanced learner representative.

Study Setup: Participants completed the study in a controlled setting with the system developer present to observe interactions and assist with setup if necessary. The AR environment was displayed on a tablet placed near the physical board, while the web interface was accessed on a laptop.

Procedure: Each participant completed two tasks in sequence.

1) **Solving a Predefined Puzzle:** Participants were first asked to solve a predefined puzzle using the command sequencer. After completing the puzzle, they were asked to solve it again using hand gesture control mode. This task was designed to compare user experiences across the two programming modalities. 2) **Puzzle Authoring and Self-Solving:** Participants were then asked to create their own puzzle using the web-based puzzle editor. After designing the puzzle, they were instructed to solve it themselves using the system. This task aimed to explore how learners engage with puzzle authoring as a form of computational thinking.

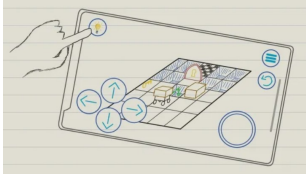
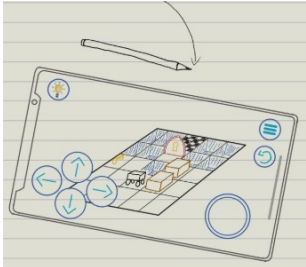
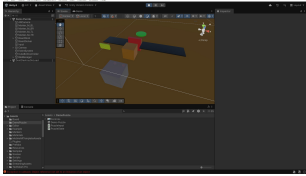
Data Collection: We collected data from written feedback forms, three open-ended questions, and reliability logs.

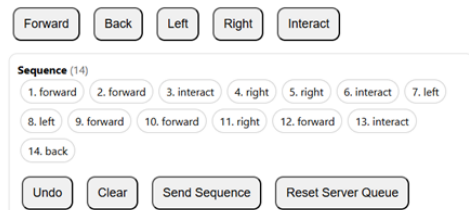
5 Key Findings and Contribution

Our formative study surfaced two central findings of a shared-state, multi-modal workflow: 1) *Sequencing supported efficient planning, while gesture control exposed reliability and feedback gaps:* Participants completed the predefined puzzle quickly with the command sequencer and characterized it as intuitive for planning and reasoning about robot actions; P1 remarked that it “*worked well*.” In contrast, hand-gesture control was consistently perceived as harder to use. Both participants reported recognition unreliability, rigid execution constraints, and limited feedback confirming whether a gesture had been correctly interpreted. Consequently, solving the same puzzle via gestures took substantially longer and produced frustration, underscoring the need for more robust recognition and clearer state-coupled feedback in gesture-based execution. 2) *Puzzle authoring was engaging but required scaffolding to translate freedom into meaningful challenges:* Participants valued the puzzle editor’s flexibility and creative potential, suggesting it could be motivating for younger learners. At the same time, the openness of the design space led to uncertainty about what constitutes a “good” or appropriately challenging puzzle. This tension indicates that learner-authored environment design can be compelling, but benefits from guidance (e.g., templates, prompts, or constraints) to help learners construct puzzles that meaningfully exercise CT concepts such as abstraction and decomposition.

Our main contributions are:

Table 1: Design iterations leading to the final AR robotic puzzle.

#	Core Idea	Image	Key Features	Feedback / Outcome
1	AR-Guided Robot Puzzle Board		Physical robot on a grid-based board; AR objects (box, key, gate) visualized on a tablet; button-based controls for movement and interaction	Critique from two researchers highlighted clear task flow and engagement, but limited support for programming concepts beyond step-by-step control
2	Embodied Control with Everyday Objects		Gesture and object-based control (e.g., pencil orientation and placement) used to command the robot; AR used to visualize puzzle state and hints	Exploration revealed strong embodied interaction potential, but state tracking and interaction complexity were difficult to scale reliably with the AR systems available
3	Programmable Logic and State Representation		Introduction of explicit puzzle state (e.g., carrying objects, unlocking gates); handling sequencing and interaction rules	Reflection indicated that mixed AR environments alone lacked sufficient novelty, motivating a redesign focused on multi-modal programming interaction and learner-authored puzzle creation
4	Multi-Modal Mixed-Reality Puzzle System		Physical robot synchronized with an AR puzzle environment; web-based interface supporting command sequencing, hand gesture control, and learner-authored puzzle design	Selected as final design; integrates multi-modal programming and learner-authored computational environments within a unified system

Cutebot Sequence BuilderBuild a sequence, then press **Send Sequence**. The commands will be executed one-by-one.**Hand Gesture Controller****Camera****Status**Detected: **back**

Camera started. Show your hand.

Start camera Stop

Default mapping:

- Open palm = forward
- One finger = interact
- Two fingers = left
- Three fingers = right
- Fist = back

Figure 2: The web-based interface supports two complementary interaction modes. The command sequencer enables symbolic, step-by-step programming of the robot (left), while the hand gesture control mode allows direct, embodied control of robot actions (right).

- A gesture-controlled AR robotic puzzle platform that supports fluid transitions between symbolic sequencing, embodied hand-gesture control, and learner-authored puzzle construction under a shared computational state.
- Early formative evidence highlighting the trade-offs between efficient planning via sequencing and the reliability/feedback

demands of gesture-based control, informing future design of robust embodied CT interfaces.

6 Future Work

Our formative evaluation identified several opportunities to refine the system and inform future research. **Improving Gesture-Based**

Interaction: Although hand gesture control was conceptually appealing, it suffered from reliability and ambiguity during use. Our future work will focus on improving gesture recognition robustness by exploring combining gesture recognition with complementary feedback, such as visual indicators or short delays before execution, to balance the immediacy of embodied interaction with the need for control and predictability. **Guided Learner-Authoring Puzzle Design:** While participants appreciated the creative freedom of the puzzle editor, excessive openness led to uncertainty and overly simple puzzle designs. Future work will explore guided mechanisms, including design prompts, example templates, constraint-based modes, and difficulty tiers aligned with specific computational thinking concepts. These mechanisms aim to balance creativity with guidance, enabling learners to construct puzzles that more meaningfully engage abstraction and decomposition. **Incorporating Local Cultural Context:** We will also explore the inclusion of puzzles and narratives that reflect the local Qatari cultural context of learners. As the system is being developed and deployed in Qatar, incorporating culturally relevant themes, such as familiar landmarks, everyday scenarios, or locally meaningful challenges, may help learners more easily relate to the puzzles and increase engagement.

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